

# CDA 3103 Computer Organization Homework #1

## 1. Fill in blanks

- 1) We study computer **organization** to become familiar with how circuits and signals collaborate to create working computer systems.
- 2) Computer **architecture** concerns itself with instruction sets and formats, operation codes, data types, the number, and types of registers, addressing modes, main memory access methods, and various I/O mechanisms.
- 3) The **instruction set architecture** is the agreed-upon interface between all the software that runs on the machine and the hardware that executes it. It allows you to talk to the machine.
- 4) Clock speeds in today's microprocessors are measured in **gigahertz**.
- 5) The computer component that makes sure that instructions are decoded and executed properly is the **control unit**.
- 6) The von Neumann bottleneck is a **single data path between the CPU and the main memory**.
- 7) At the most basic level, a computer is a device consisting of three pieces: A processor; an I/O mechanism and a(n) **memory**.
- 8) In the von Neumann architecture, the central processing unit (CPU) consists of registers, an arithmetic-logic unit (ALU), and a(n) **control unit**.
- 9) Expressed as a power of two, there are **2<sup>10</sup>** kilobytes in a megabyte.
- 10) Expressed as a power of two, there are **2<sup>20</sup>** megabytes in a terabyte.
- 11) A terabyte represents what power of two? **40**.
- 12) A kilobyte represents what power of two? **10**.
- 13) A gigabyte represents what power of two? **30**.
- 14) In 1965, one of the founders of Intel predicted "The density of transistors in an integrated circuit will double every year." This is now known as: **Moore's Law**.
- 15) The statement "The cost of capital equipment to build semiconductors will double every four years." is known as: **Rock's Law**.

## 2. Explain why modern computers consist of multiple abstract levels and describe the main functions of each level.

Ans: This allows us to use a split and trounce approach and view the computer organization as separate layers, each built upon the ones below it. The different layers are as follows:

**Level 6 (User Level):** This is an interface where the user runs and executes programs.

**Level 5 (High Level Language):** In this, we use compiler or interpreter to translate high level languages to assembly language which is then assembled into machine code.

**Level 4 (Assembly Language Level):** This level compresses the assembly language. The compilers change the high level language to machine level languages.

**Level 3 (System Software Level):** This level deals with Operating system and it is responsible for tasks such as multiprogramming protecting memory, synchronizing process etc. Code translated from assembly language to machine language is passed unmodified through this level.

**Level 2 (Machine Level):** The code which is translated into machine code is executed directly by electronic circuits without any interpreters and translators.

**Level 1 (Control Level):** This level ensures that all the instructions are executed and decoded properly, and the data is moved at appropriate place and time.

**Level 0 (Digital Logic Level):** This is the level of computer hierarchy that is composed of physical components, such as gates and wires.

3. In the von Neumann model, explain the purpose of the following components:
  - a) **Central processing unit** – CPU is a part of the computer that performs the processing of information. CPU can also be understood as an electric circuit that is used to process instructions of a computer program. The CPU performs various types of information.
  - b) **Program counter** – Program counter is a register that manages the memory addresses of the instruction to be executed next. As each instruction gets fetched, the program counter increases its stored value by 1.
  - c) **Arithmetic logic unit** – ALU is a digital circuit and a basic block of the CPU. The arithmetic unit is responsible for operations like + and – whereas the logic unit is responsible for operands like AND and NOT.
  - d) **Control unit** – Control unit is a part of the CPU that directs the operation of the processes. It directs the flow of data between the CPU and the other units.

4. Explain what it means to “fetch” an instruction.

Ans: The term fetch means that some information is loaded from the memory to CPU registers. Fetching the information is important because without that the information will not be executed.

5. Explain what it means to “decode” an instruction.

Ans: Decoding is a process in which the CPU determines which instructions are necessary to perform the operation. The opcode fetched from the memory is decoded for the next steps and moved to the appropriate registers.

6. Describes the major differences between computer organization and computer architecture.

Ans: Computer Architecture is concerned with the way hardware components are connected to form a computer system. Computer Organization is concerned with the structure and behavior of a computer system as seen by the user. It acts as the interface between hardware and software.

7. How does the fetch-decode-execute cycle work?

Ans: The mentioned cycle is the instruction cycle which is CPU follows for fetching, decoding, and executing the instructions. The cycle goes as the following:

**Fetch Stage:** This refers to the fetching of the next instruction from the memory which is stored with the program counter and then is stored in the instruction register.

**Decode Stage:** It is the stage when the decoder will understand the encoded instruction that is being fetched and that is already in the instruction register.

**Execute Stage:** This is the stage where the control unit passes the decoded information for the instruction as the control signals to relevant CPU units for performing the instruction execution, and thus ALU operations are performed here, and feedback is passed.